

Murat Khidoyatov

U.S. Permanent Resident

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Education

Yale University

Graduation Date: May 2027

B.S. Mechanical Engineering (ABET)

Relevant Coursework: Programming in C/C++, Linear Algebra, Differential Equations, Fluid Mechanics, Static, Dynamics, Computational Mechanics

Awards: Competitive Yale personal research grant (8% acceptance rate) · Yale First-Year Research Fellowship

Technical Skills

Languages & Frameworks: Python, C/C++, JAX, PyTorch, NumPy, CUDA, OPTAX

Simulation & Scientific Computing: CFD (Python), FEA (COMSOL, ABAQUS), Numerical Methods, Automatic Differentiation

AI/ML: Physics-Informed Neural Networks, Energy-Based Models, Bayesian Optimization, Random Forest, k-NN

HPC & Tools: GPU Acceleration (NVIDIA L4, A100), MATLAB, Linux, HPC

Research Experience

Yale Engineering MechanoBiology Lab

New Haven, CT · 2023 – Present

Undergraduate Researcher

- Developed CFD simulations of flow around a cylinder in Python, generating ground-truth flow field datasets used to train and validate physics-informed neural networks.
- Built a Physics-Informed Neural Network (PINN) in JAX enforcing fluid and structural governing equations via automatic differentiation, achieving 95% accuracy relative to finite element solutions.
- Accelerated neural network training by integrating CUDA-based GPU kernels, reducing training time on NVIDIA L4 and A100 GPUs
- Built a dual-network framework coupling displacement prediction with a learned energy function, achieving $\pm 90\%$ accuracy for Neo-Hookean non-linear elasticity vs. COMSOL benchmarks.
- Co-authored pending publication on physics-consistent neural networks for deformation and director field learning in microstructured media.

Oak Ridge National Laboratory

Oak Ridge, TN · Summer 2025

Research Intern

- Built ML surrogate models on experimental datasets achieving 96% predictive accuracy, integrated with Bayesian Optimization to navigate high-dimensional alloy design spaces near global optima.
- Developed a full Bayesian Optimization pipeline in Python combining surrogate inference with acquisition functions to efficiently explore sparse, noisy experimental data near global optima.
- Interpreted learned models via feature-response correlation analysis to validate physically meaningful solutions; presented findings as a research poster on ML-driven material optimization.

Publication

Shirani, M., Gueldner, P. H., Khidoyatov, M., Warren, J. L., Ninno, F., & Humphrey, J. D. (2026). Physics-Consistent Neural Networks for Learning Deformation and Director Fields in Microstructured Media with Loss-Based Validation Criteria. *(To Be Submitted)*

Project Experience

Yale Undergraduate Robotics

2024 – Present

President & Science Sub-team Lead

- Programmed a forward and inverse kinematic simulator in Python for real-time robotic arm path planning and joint validation.
- Developed a custom spectrometer using image analysis and Fourier decomposition of light wave data.
- Led a 10-member team designing mechanical and software systems for a Mars rover life-detection system.